



Car Dataset Analysis

data .csv <https://drive.google.com/file/d/19V7VjHs4J8idTko7NQUT95ZJO2ZQSGWh/view>

[Car Data Analysis_Adriana.ipynb](#)

 **README**

This project focused on using Python for data cleaning, feature engineering, and exploratory analysis of a car dataset.

List of Columns in DataFrame

1. **Make:** The manufacturer of the car (e.g., Toyota, Ford, BMW).
2. **Model:** The specific model name of the car.
3. **Year:** The production year of the car.
4. **Engine Fuel Type:** Type of fuel the car uses (e.g., gas, diesel, electric, hybrid).
5. **Engine HP:** The horsepower of the car's engine.
6. **Engine Cylinders:** The number of cylinders in the car's engine.
7. **Transmission Type:** The type of transmission in the car (e.g., automatic, manual).
8. **Driven_Wheels:** The type of drivetrain (e.g., front-wheel drive, rear-wheel drive, all-wheel drive).
9. **Number of Doors:** Number of doors the car has (e.g., 2, 4).
10. **Market Category:** The market segment the car belongs to (e.g., SUV, sedan, luxury).
11. **Vehicle Size:** The size category of the car (e.g., compact, mid-size, large).
12. **Vehicle Style:** The style of the vehicle (e.g., coupe, convertible, hatchback).
13. **highway MPG:** The estimated miles per gallon on the highway.
14. **city mpg:** The estimated miles per gallon in the city.
15. **Popularity:** A numerical value indicating how popular the car is in the market.
16. **MSRP:** The Manufacturer's Suggested Retail Price of the car.

Project Goal

The project includes 3 main tasks, each with their subtasks:

Task 1. Data Cleansing

- filtering for cars from **1995**
- handling missing values,
- ensuring correct data types,
- standardizing text entries in specific columns.

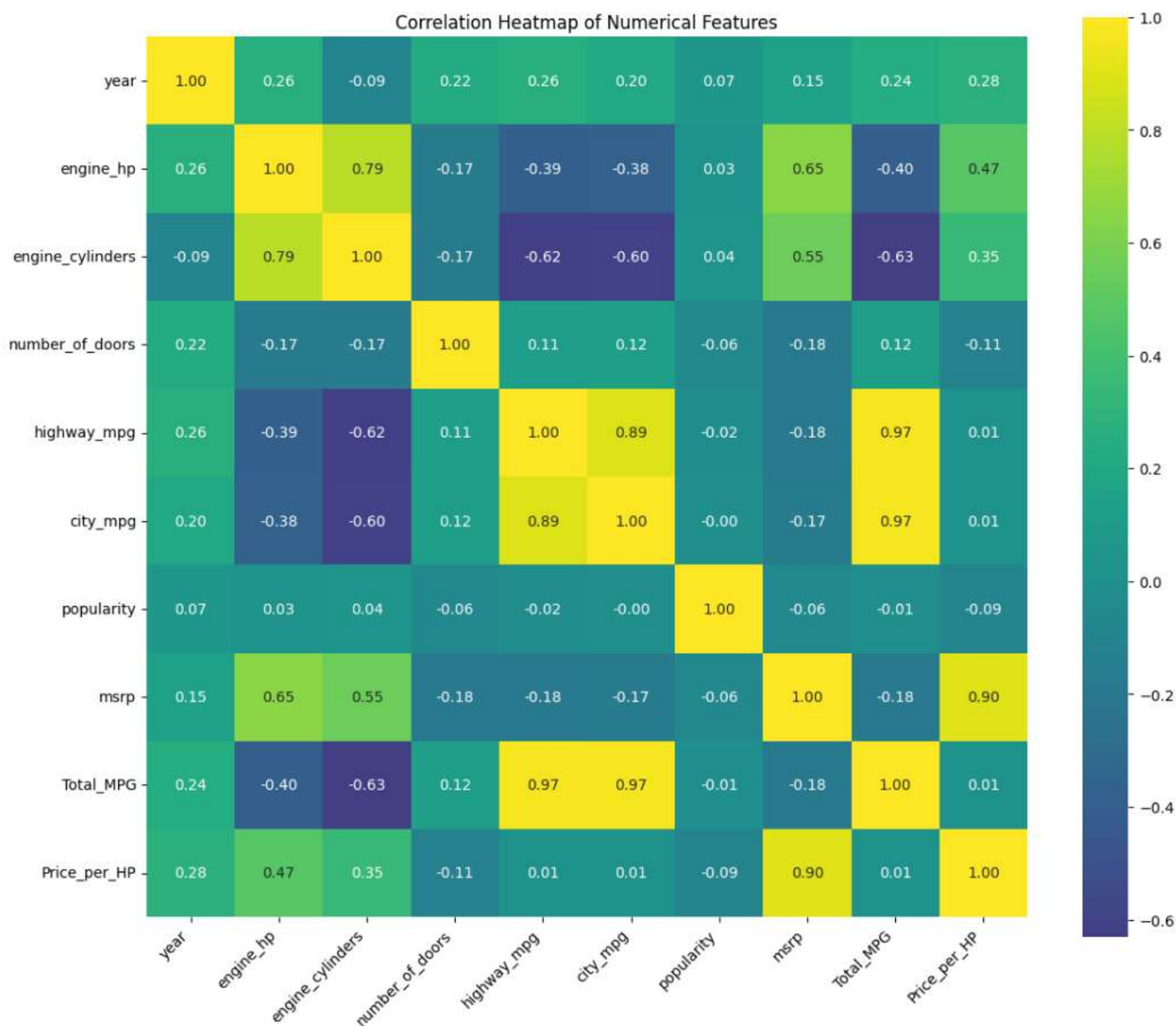
Task 2. Feature Engineering

Second task is to create new columns for analysis:

1. calculate the average of city and highway MPG as **Total MPG**
2. determine **Price per HP** by dividing MSRP by Engine HP.

Task 3. Exploratory Data Analysis (EDA)

The third task involves exploring the dataset using **descriptive statistics**, group-based analysis, and various **visualizations** to uncover patterns. Additionally, perform a correlation analysis to understand relationships between key variables.



legend of colored squares:

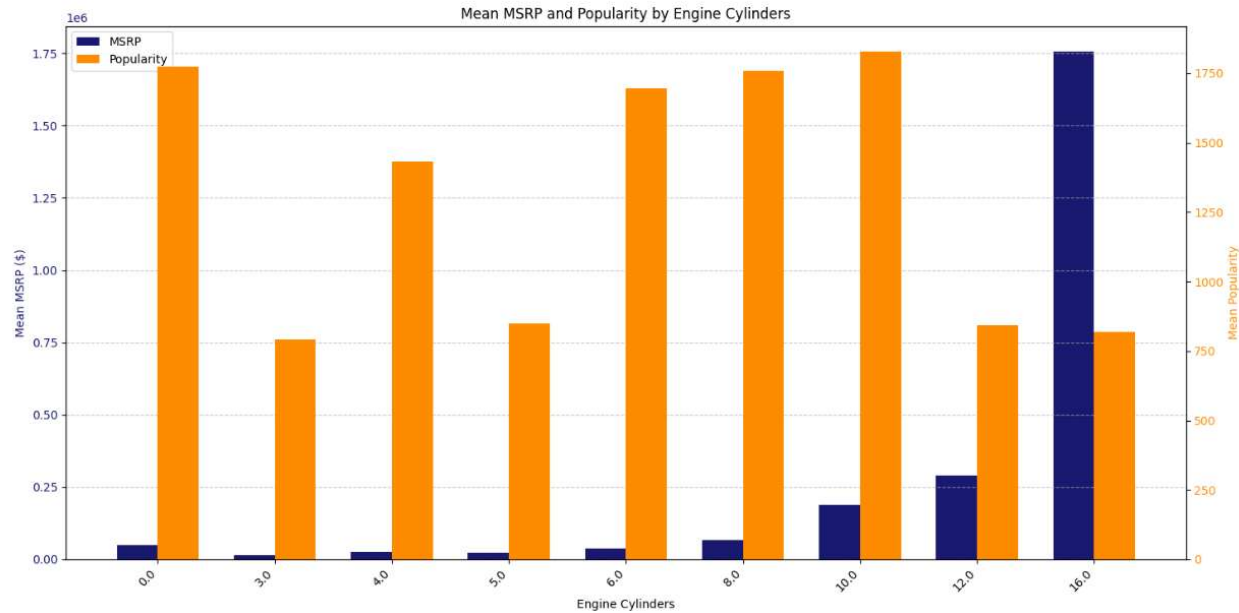
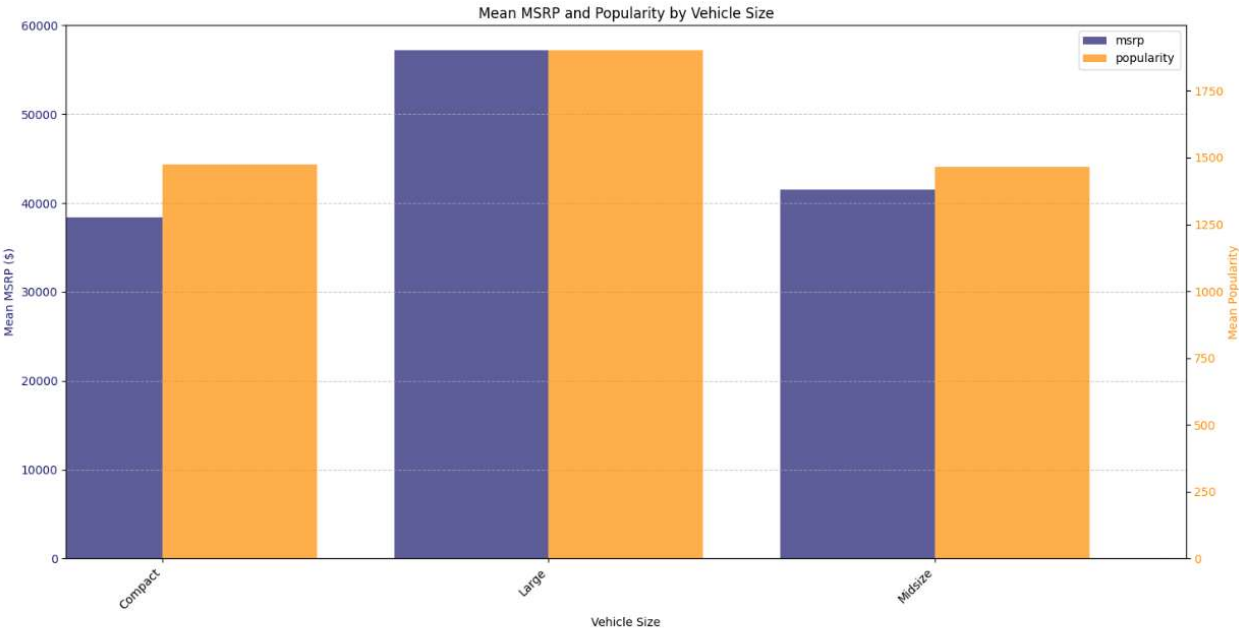
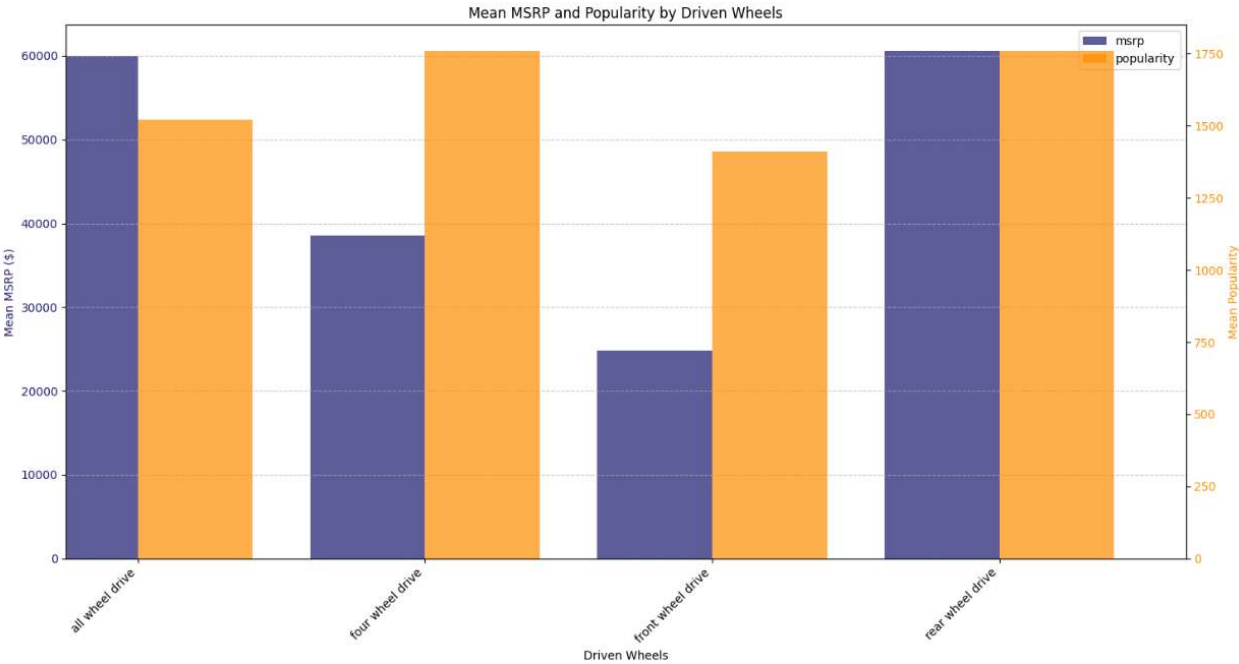
- **yellow**(ex.: Total_MPG with city_mpg or highway / Price_per_HP with msrp)
and
- **dark blue** (ex.: Total_MPG with engine_cylinders or engine_hp / engine_cylinders with city_mpg or highway_mpg) show a **strong linear correlation** between the numerical features
- **green** (ex.: popularity with city_mpg or highway_mpg, or msrp) indicates a **weak or nonlinear correlation** between the numerical features

Mean *MSRP* and *Popularity* for each group:

Driven_Wheels

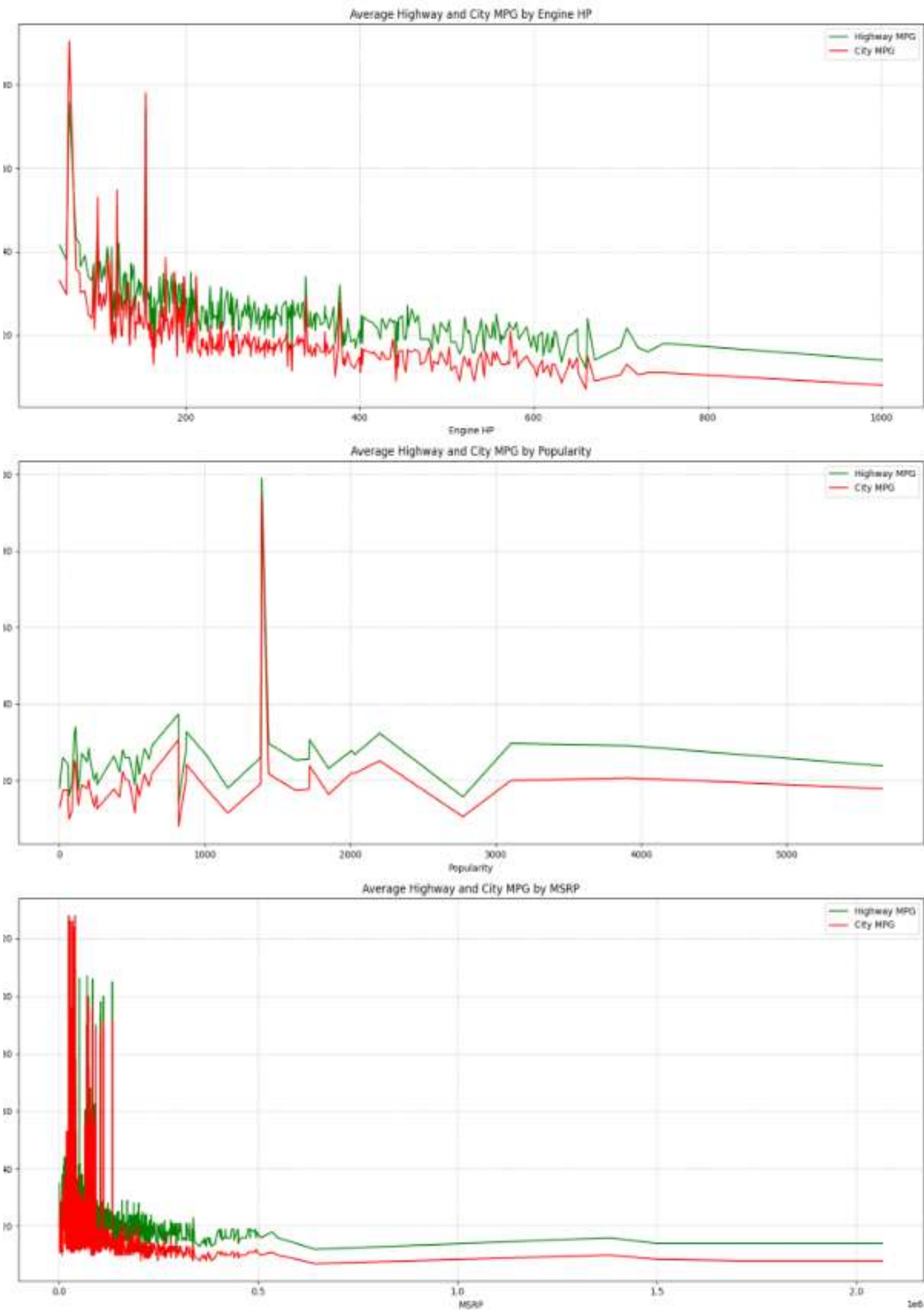
Vehicle Size

Engine Cylinders

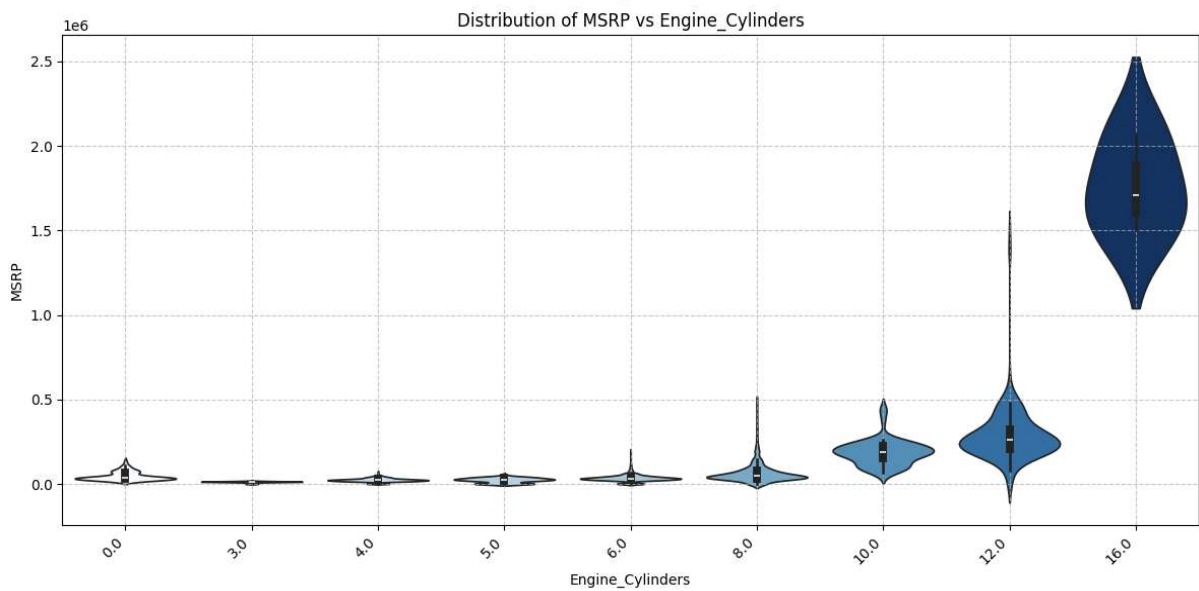
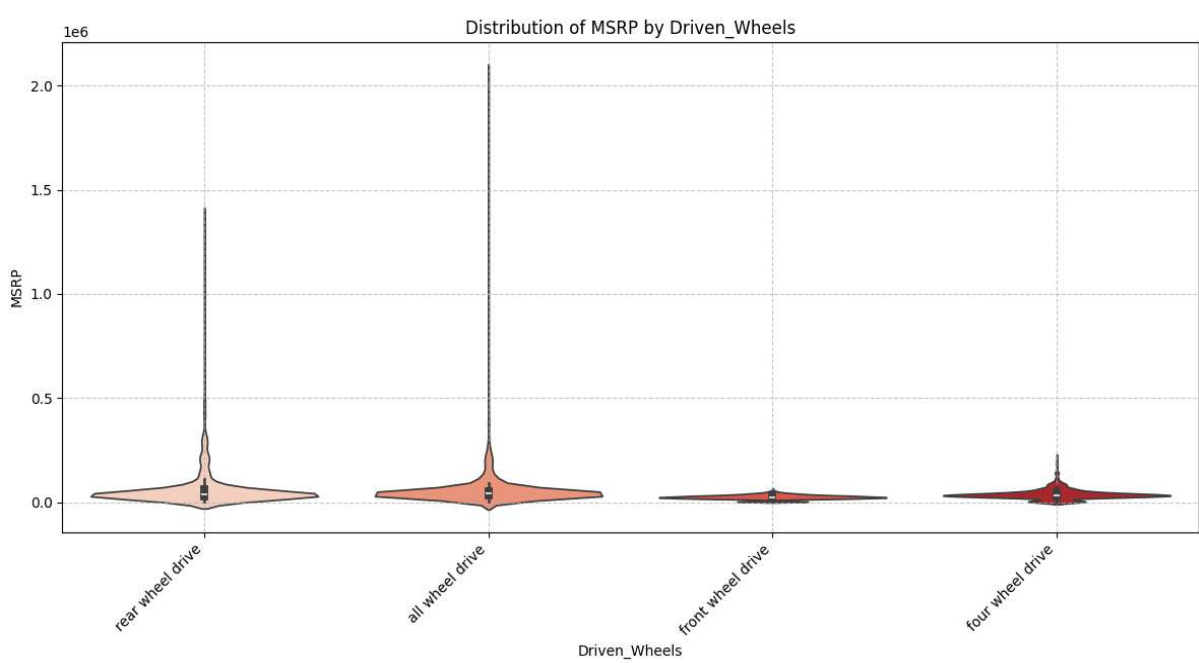
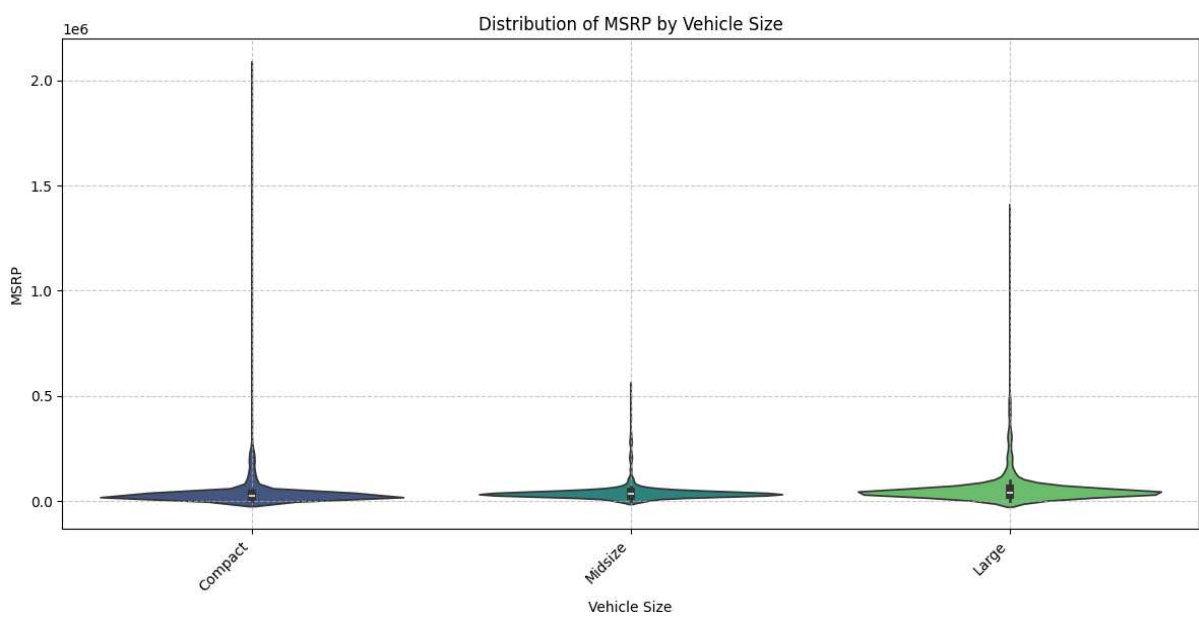


06.2

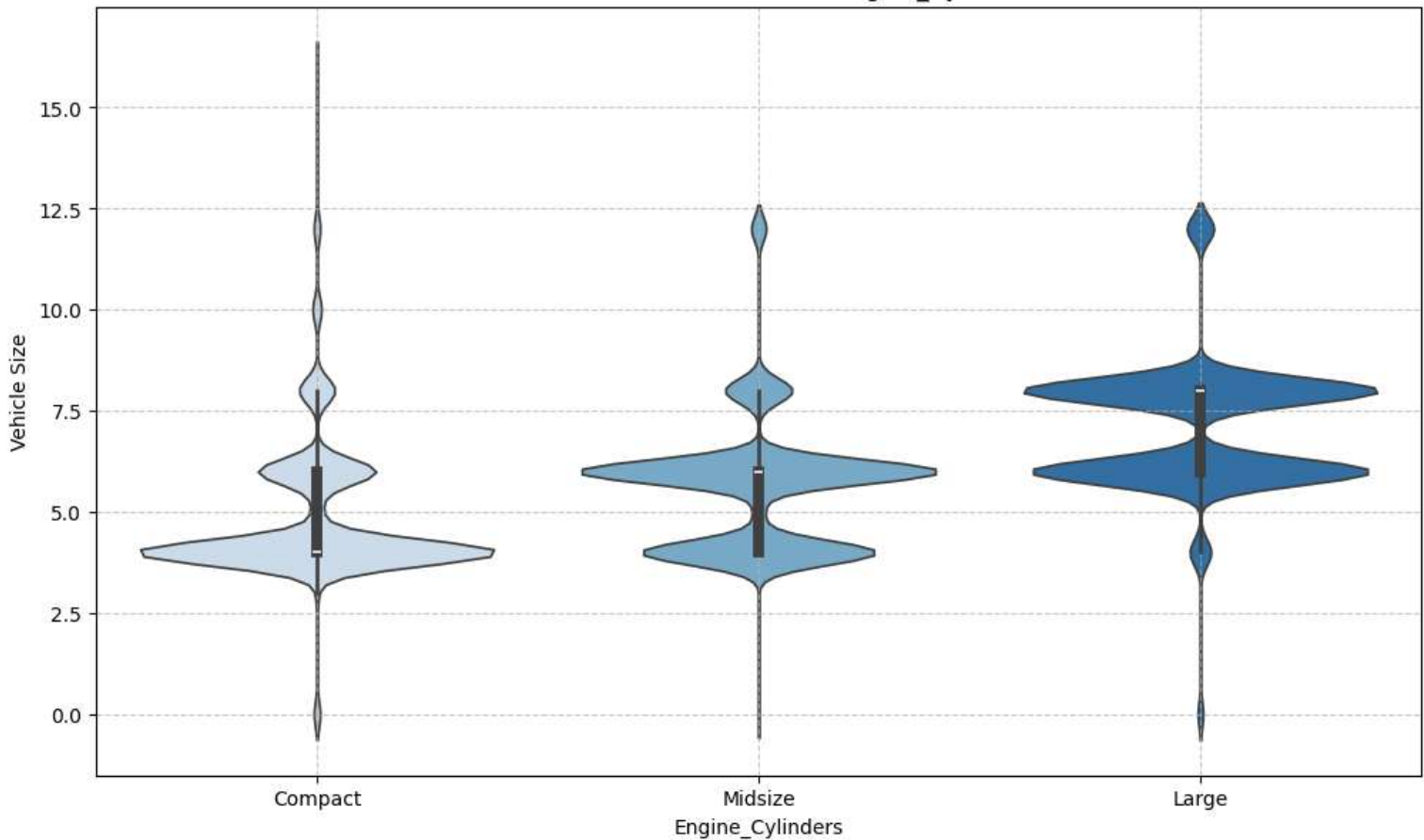
Clear comparison of fuel efficiency against these variables.



Violin Plots (Distribution, Correlation , Outliers)



Distribution of Vehicle Size vs Engine_Cylinders



Midsize Vehicles analysis

The **Midsize vehicle category** exhibits characteristics that generally fall between 'Compact' and 'Large' vehicles across several key metrics:

- **Mean MSRP****: Approximately ****\$41,505.44**** (as seen in ``df_grouped_vehicle_size``).

This positions 'Midsize' vehicles squarely between 'Compact' (mean MSRP ~\$38,408) and 'Large' vehicles (mean MSRP ~\$57,177).

- **Mean Popularity****: Around ****1,464.65**** (from ``df_grouped_vehicle_size``).

Similar to MSRP, its popularity score is intermediate, not the highest nor the lowest.

- **Mean Engine Cylinders****: Averaging ****5.30**** cylinders (from ``df_grouped_engine_cylinders``).

This is higher than 'Compact' vehicles (mean ~4.57) but lower than 'Large' vehicles (mean ~6.86).

Visual Confirmations from Violin Plots:

- **Distribution of MSRP by Vehicle Size****: The violin plot for 'Midsize' vehicles shows a distribution of MSRPs that is wider than 'Compact' but often less spread out at the highest prices than 'Large', confirming its intermediate pricing range.
- **Distribution of Vehicle Size vs Engine Cylinders****: The 'Midsize' violin plot primarily shows concentrations around 4 and 6 cylinders, with some representation of 8-cylinder engines. This visually aligns with the calculated mean of 5.30 cylinders.

Outlier Analysis. Observations for Midsize Vehicles:

The analysis identified:

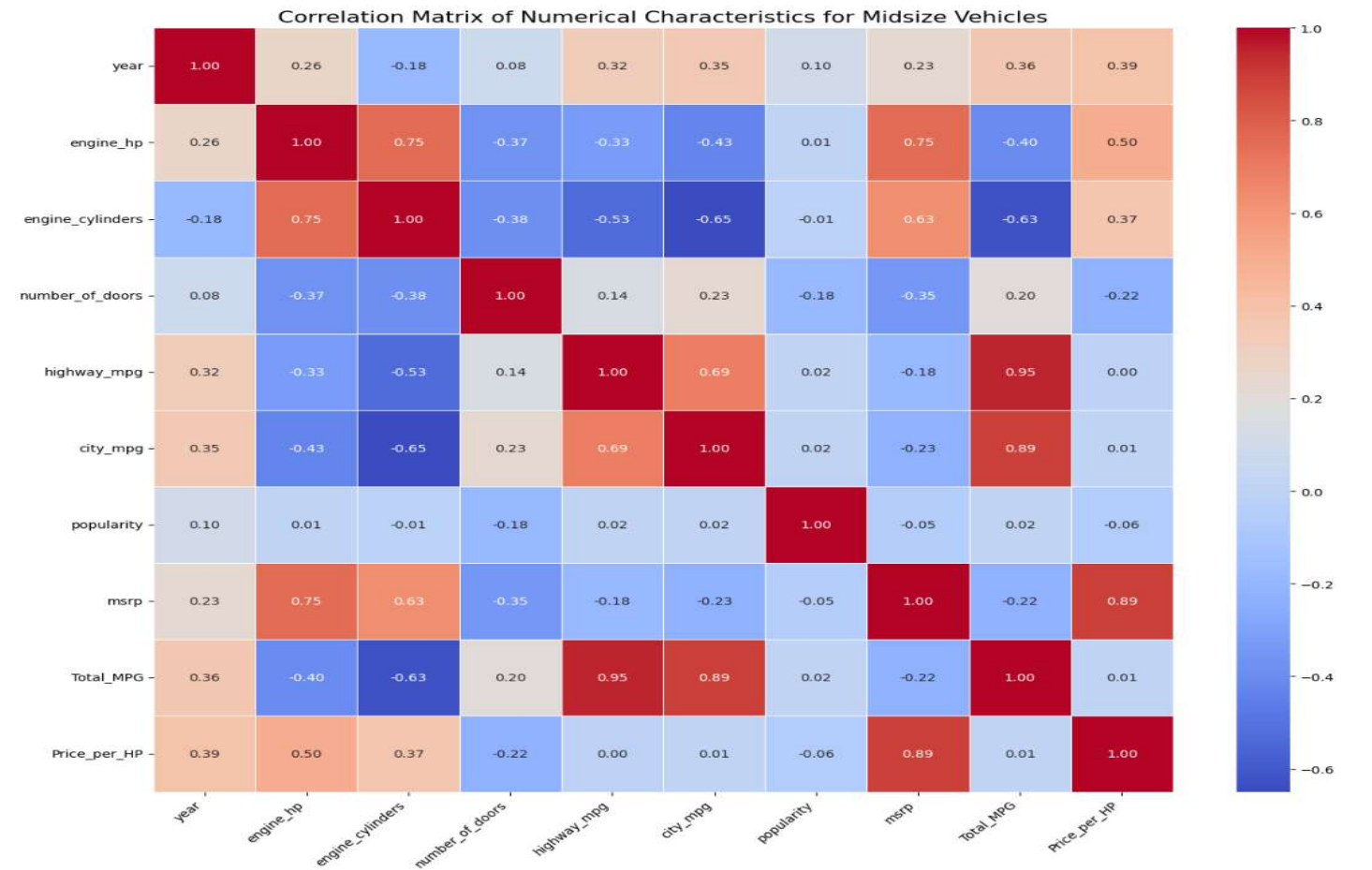
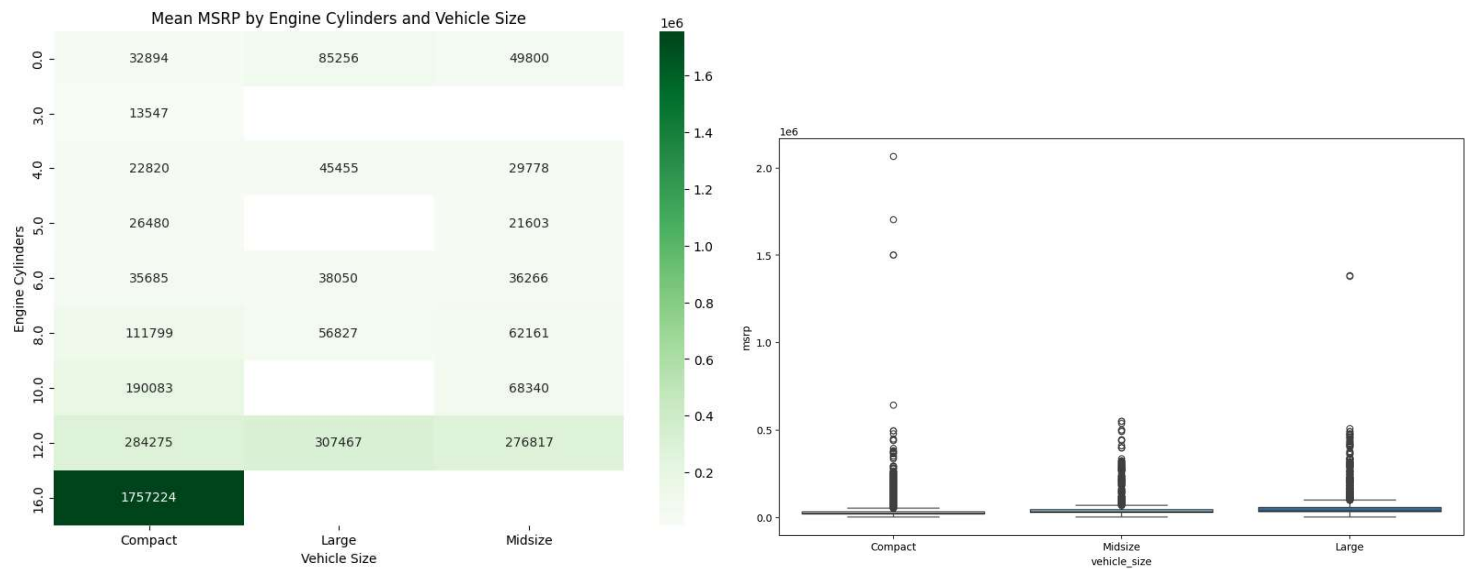
- **252 MSRP outliers** in the 'Midsize' category, and an **upper bound of approximately \$68,898.75**.

This indicates that while the bulk of midsize vehicles fall within a certain price range, there are a **significant number of models priced considerably higher, some reaching up to \$548,800.00**.

- **89 engine cylinder outliers** in 'Midsize' vehicles.

Notably, **these include vehicles with 0 cylinders** (electric cars, which have no traditional engine cylinders) and a **significant number of vehicles with 12 cylinders**.

This suggests that while most midsize cars use 4 or 6 cylinders, a **small segment includes electric vehicles and high-performance luxury models with many cylinders**.



SUMMARY

of the 'Midsize' vehicle category, which is derived from our *cleaned and filtered dataset*, highlights several key characteristics:

Midsize Vehicle Characteristics (based on cleaned data):

- **Mean MSRP:** Approximately **\$41,505.44**. This positions Midsize vehicles robustly between 'Compact' (mean MSRP ~\$38,408) and 'Large' (mean MSRP ~\$57,177) categories. This figure is more reliable due to the prior handling of MSRP outliers.
- **Mean Popularity:** Around **1,464.65**. Similar to MSRP, its popularity score is intermediate, providing a balanced view after removing duplicate entries.
- **Mean Engine Cylinders:** Averaging **5.30** cylinders. This is higher than 'Compact' vehicles (mean ~4.57) but lower than 'Large' vehicles (mean ~6.86). The accuracy of this average is improved by addressing missing `engine_cylinders` values, especially for electric cars and specific manufacturers like Mazda.

Impact of Filtering and Cleaning on these Insights:

Our analysis benefits significantly from the preceding data wrangling steps:

1. **Column Name Standardization:** Lowercasing and replacing spaces with underscores in column names ensured consistency for analysis.
2. **Year Filtering:** Removing vehicles older than 1995 focused the dataset on more modern and relevant vehicles, preventing historical outliers from skewing averages.
3. **Duplicate Removal:** Dropping 487 duplicate rows ensured that each unique vehicle entry contributed fairly to the statistical calculations, preventing overrepresentation.
4. **Handling Missing Values:**
 - `engine_fuel_type` missing values were imputed based on context .
 - `engine_hp` missing values were carefully replaced using contextual information (e.g., median for specific makes/types) rather than simple global imputation . This ensures that the engine performance statistics for Midsize vehicles are as accurate as possible.
 - `engine_cylinders` missing values were specifically addressed, assigning 0 for electric vehicles and appropriate values for others like Mazda. This is crucial for accurate engine-related analysis within the Midsize category, especially when identifying outliers.
 - `number_of_doors` missing values were also imputed .

These cleaning efforts directly confirm the reality of the data presented for the 'Midsize' category, ensuring that the statistics and observations accurately reflect the characteristics of these vehicles within our refined dataset.

Choosing the subject of a future analysis is significantly easier, using **Pairplot** (It helps to observe correlations, patterns, and potential clusters between all numerical features).

Pairwise Relationships of All Numerical Features in Midsize Vehicles

